Scalability & Future Plan

Date	30 September 2026
Team ID	SWTID-2026-3818
Project Name	Auto_Ticket_Classification_Using_Flow_Diagram
Maximum Marks	1 Mark

This document explains how the solution can be scaled to handle more users, more data and more features, and gives the team's roadmap.

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	SQLite handles limited concurrent writes	Slower when many users submit tickets together	Medium
2	Classical ML model relies on keywords (TF-IDF)	May miss the meaning of long or unusual tickets	High
3	Supports English text only	Cannot classify tickets in other languages	Medium
4	Fixed set of categories	New issue types need retraining	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Flask development server on one machine	Gunicorn + Nginx, container (Docker), cloud hosting with load balancing
2	Data Storage	SQLite file	PostgreSQL / MySQL with regular backups
3	Performance	Model loaded once in memory; one request at a time	Caching, batch prediction, background workers (Celery)
4	Security	Basic login and input validation	HTTPS, JWT tokens, rate limiting, audit logs

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Use a transformer model (DistilBERT) for better accuracy	Short term (1-3 months)	Higher accuracy on complex tickets
Phase 2	Email and Jira / ServiceNow integration to receive tickets automatically	Short term (3-4 months)	No manual entry
Phase 3	Multilingual ticket support	Medium term (4-6 months)	Wider user base
Phase 4	Chatbot that collects ticket details step by step	Medium term (6 months)	Better quality tickets
Phase 5	Suggested solutions using a generative AI model	Long term (6-12 months)	Faster resolution